

Supplementary results: curated resistance gene families and Evo 2

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3 October 2026

All values below were checked against the original completed-run predictions. EA is essential agreement against an interval-censored MIC reference; EA exact uses exact references only. Brackets give 95% lineage-bootstrap intervals (2,000 draws, fixed predictions). Models with family features are post hoc. No model fitting was performed during manuscript verification.

S1. Locked external evaluation and exploratory family models

Human clinical

Ceftriaxone

All models: n = 1505; exact-reference n = 359.

Model	EA [95% interval]	EA exact
Constant	0.290 [0.121, 0.545]	0.000
Nearest relative	0.529 [0.415, 0.696]	0.390
Ridge: allele	0.680 [0.539, 0.742]	0.237
Ridge: allele + Evo 2	0.684 [0.546, 0.748]	0.315
Ridge: Evo 2	0.579 [0.465, 0.680]	0.217
XGBoost: allele	0.597 [0.480, 0.692]	0.365
XGBoost: allele + Evo 2	0.612 [0.462, 0.695]	0.421
XGBoost: Evo 2	0.541 [0.419, 0.649]	0.415
Ridge: allele + family + class*	0.771 [0.666, 0.817]	0.365
XGBoost: allele + family + class*	0.854 [0.784, 0.892]	0.652

Family models were developed after inspection of the original external results.

Ciprofloxacin

All models: n = 1505; exact-reference n = 260.

Model	EA [95% interval]	EA exact
Constant	0.400 [0.213, 0.653]	0.000
Nearest relative	0.849 [0.769, 0.899]	0.746
Ridge: allele	0.932 [0.894, 0.950]	0.681
Ridge: allele + Evo 2	0.908 [0.866, 0.929]	0.662
Ridge: Evo 2	0.870 [0.820, 0.911]	0.419
XGBoost: allele	0.933 [0.913, 0.954]	0.685
XGBoost: allele + Evo 2	0.935 [0.911, 0.955]	0.708
XGBoost: Evo 2	0.923 [0.890, 0.943]	0.650
Ridge: allele + family + class*	0.904 [0.838, 0.948]	0.496
XGBoost: allele + family + class*	0.938 [0.918, 0.959]	0.719

Family models were developed after inspection of the original external results.

Gentamicin

All models: n = 1505; exact-reference n = 184.

Model	EA [95% interval]	EA exact
Constant	0.783 [0.703, 0.883]	0.755
Nearest relative	0.752 [0.667, 0.877]	0.761
Ridge: allele	0.887 [0.841, 0.916]	0.239
Ridge: allele + Evo 2	0.854 [0.812, 0.873]	0.087
Ridge: Evo 2	0.822 [0.777, 0.847]	0.065
XGBoost: allele	0.876 [0.829, 0.909]	0.228
XGBoost: allele + Evo 2	0.856 [0.823, 0.873]	0.087
XGBoost: Evo 2	0.824 [0.773, 0.848]	0.092
Ridge: allele + family + class*	0.880 [0.855, 0.904]	0.179
XGBoost: allele + family + class*	0.901 [0.871, 0.928]	0.277

Family models were developed after inspection of the original external results.

All sources

Ceftriaxone

All models: n = 1505; exact-reference n = 359.

Model	EA [95% interval]	EA exact
Constant	0.290 [0.121, 0.545]	0.000
Nearest relative	0.510 [0.383, 0.671]	0.345
Ridge: allele	0.705 [0.598, 0.759]	0.362
Ridge: allele + Evo 2	0.701 [0.579, 0.756]	0.443
Ridge: Evo 2	0.627 [0.502, 0.696]	0.220
XGBoost: allele	0.696 [0.593, 0.756]	0.582
XGBoost: allele + Evo 2	0.678 [0.523, 0.746]	0.518
XGBoost: Evo 2	0.583 [0.413, 0.686]	0.457

Family models were developed after inspection of the original external results.

Ciprofloxacin

All models: n = 1505; exact-reference n = 260.

Model	EA [95% interval]	EA exact
Constant	0.400 [0.213, 0.653]	0.000
Nearest relative	0.847 [0.777, 0.894]	0.665
Ridge: allele	0.938 [0.888, 0.959]	0.692
Ridge: allele + Evo 2	0.936 [0.911, 0.953]	0.773
Ridge: Evo 2	0.900 [0.863, 0.923]	0.588
XGBoost: allele	0.944 [0.917, 0.960]	0.723
XGBoost: allele + Evo 2	0.947 [0.924, 0.959]	0.762
XGBoost: Evo 2	0.928 [0.879, 0.949]	0.673

Family models were developed after inspection of the original external results.

Gentamicin

All models: n = 1505; exact-reference n = 184.

Model	EA [95% interval]	EA exact
Constant	0.783 [0.703, 0.883]	0.755
Nearest relative	0.728 [0.645, 0.846]	0.582
Ridge: allele	0.890 [0.870, 0.905]	0.217
Ridge: allele + Evo 2	0.889 [0.869, 0.904]	0.217
Ridge: Evo 2	0.817 [0.766, 0.850]	0.174
XGBoost: allele	0.880 [0.862, 0.899]	0.120
XGBoost: allele + Evo 2	0.877 [0.862, 0.895]	0.092
XGBoost: Evo 2	0.839 [0.802, 0.860]	0.098

Family models were developed after inspection of the original external results.

S2. Final development out-of-fold evaluations

Human clinical

Ceftriaxone

All models: n = 2822; exact-reference n = 157.

Model	EA [95% interval]	EA exact
Constant	0.721 [0.583, 0.866]	0.000
Nearest relative	0.692 [0.561, 0.833]	0.057
Ridge: allele	0.918 [0.889, 0.947]	0.210
Ridge: allele + Evo 2	0.926 [0.901, 0.953]	0.280
Ridge: Evo 2	0.895 [0.857, 0.936]	0.159
XGBoost: allele	0.929 [0.900, 0.957]	0.490
XGBoost: allele + Evo 2	0.914 [0.884, 0.946]	0.484
XGBoost: Evo 2	0.888 [0.851, 0.931]	0.401
Ridge: allele + family + class*	0.924 [0.900, 0.952]	0.261
XGBoost: allele + family + class*	0.933 [0.898, 0.964]	0.535

Family models were developed after inspection of the original external results.

Ciprofloxacin

All models: n = 3809; exact-reference n = 647.

Model	EA [95% interval]	EA exact
Constant	0.599 [0.408, 0.826]	0.481
Nearest relative	0.566 [0.403, 0.765]	0.371
Ridge: allele	0.913 [0.891, 0.939]	0.584
Ridge: allele + Evo 2	0.910 [0.888, 0.936]	0.583
Ridge: Evo 2	0.884 [0.859, 0.911]	0.512
XGBoost: allele	0.918 [0.900, 0.940]	0.628
XGBoost: allele + Evo 2	0.913 [0.891, 0.934]	0.594
XGBoost: Evo 2	0.747 [0.537, 0.924]	0.635
Ridge: allele + family + class*	0.914 [0.894, 0.940]	0.597
XGBoost: allele + family + class*	0.917 [0.901, 0.934]	0.611

Family models were developed after inspection of the original external results.

Gentamicin

All models: n = 3348; exact-reference n = 440.

Model	EA [95% interval]	EA exact
Constant	0.823 [0.733, 0.923]	0.791
Nearest relative	0.813 [0.725, 0.909]	0.820
Ridge: allele	0.885 [0.866, 0.910]	0.343
Ridge: allele + Evo 2	0.912 [0.894, 0.931]	0.527
Ridge: Evo 2	0.882 [0.862, 0.907]	0.457
XGBoost: allele	0.894 [0.861, 0.931]	0.441
XGBoost: allele + Evo 2	0.918 [0.901, 0.941]	0.577
XGBoost: Evo 2	0.885 [0.853, 0.925]	0.577
Ridge: allele + family + class*	0.893 [0.870, 0.919]	0.345
XGBoost: allele + family + class*	0.895 [0.869, 0.927]	0.368

Family models were developed after inspection of the original external results.

All sources

Ceftriaxone

All models: n = 7097; exact-reference n = 830.

Model	EA [95% interval]	EA exact
Constant	0.767 [0.693, 0.843]	0.020
Nearest relative	0.657 [0.590, 0.726]	0.142
Ridge: allele	0.926 [0.914, 0.941]	0.559
Ridge: allele + Evo 2	0.929 [0.916, 0.944]	0.598
Ridge: Evo 2	0.871 [0.849, 0.899]	0.252
XGBoost: allele	0.942 [0.930, 0.954]	0.736
XGBoost: allele + Evo 2	0.921 [0.902, 0.940]	0.658
XGBoost: Evo 2	0.869 [0.845, 0.899]	0.452

Family models were developed after inspection of the original external results.

Ciprofloxacin

All models: n = 8493; exact-reference n = 1156.

Model	EA [95% interval]	EA exact
Constant	0.747 [0.616, 0.864]	0.398
Nearest relative	0.645 [0.543, 0.741]	0.385
Ridge: allele	0.917 [0.904, 0.931]	0.465
Ridge: allele + Evo 2	0.928 [0.916, 0.938]	0.548
Ridge: Evo 2	0.911 [0.899, 0.924]	0.437
XGBoost: allele	0.933 [0.924, 0.944]	0.587
XGBoost: allele + Evo 2	0.934 [0.924, 0.945]	0.593
XGBoost: Evo 2	0.926 [0.913, 0.941]	0.556

Family models were developed after inspection of the original external results.

Gentamicin

All models: n = 9514; exact-reference n = 5221.

Model	EA [95% interval]	EA exact
Constant	0.811 [0.763, 0.858]	0.919
Nearest relative	0.727 [0.675, 0.775]	0.759
Ridge: allele	0.926 [0.905, 0.944]	0.909
Ridge: allele + Evo 2	0.929 [0.913, 0.944]	0.914
Ridge: Evo 2	0.884 [0.863, 0.908]	0.890
XGBoost: allele	0.932 [0.912, 0.948]	0.908
XGBoost: allele + Evo 2	0.935 [0.921, 0.949]	0.910
XGBoost: Evo 2	0.897 [0.875, 0.921]	0.888

Family models were developed after inspection of the original external results.

S3. Variant withholding: all carriers and above-panel carriers

For each target, all its development carriers were withheld. EA uses all held-out carriers; the >2 mg/L share uses right-censored references only. The four primary CTX-M targets contribute 568 target-isolate observations from 565 unique isolates and 45 lineages.

blaCMY-2

Learner / features	Held out	Above panel	EA	>2 mg/L share
Ridge / allele	78	22	0.103	0.182
Ridge / allele + Evo 2	78	22	0.115	0.182
Ridge / evo2	78	22	0.179	0.227
Ridge / allele + family + class	78	22	0.487	0.773
XGBoost / allele	78	22	0.103	0.182
XGBoost / allele + Evo 2	78	22	0.231	0.318
XGBoost / evo2	78	22	0.385	0.591
XGBoost / allele + family + class	78	22	0.256	0.727

blaCTX-M-14

Learner / features	Held out	Above panel	EA	>2 mg/L share
Ridge / allele	47	37	0.043	0.054
Ridge / allele + Evo 2	47	37	0.064	0.189
Ridge / evo2	47	37	0.511	0.973
Ridge / allele + family + class	47	37	0.894	1.000
XGBoost / allele	47	37	0.064	0.027
XGBoost / allele + Evo 2	47	37	0.149	0.676
XGBoost / evo2	47	37	0.447	1.000
XGBoost / allele + family + class	47	37	0.936	1.000

blaCTX-M-15

Learner / features	Held out	Above panel	EA	>2 mg/L share
Ridge / allele	410	361	0.076	0.080

Learner / features	Held out	Above panel	EA	>2 mg/L share
Ridge / allele + Evo 2	410	361	0.076	0.235
Ridge / evo2	410	361	0.644	0.922
Ridge / allele + family + class	410	361	0.763	0.972
XGBoost / allele	410	361	0.029	0.042
XGBoost / allele + Evo 2	410	361	0.098	0.687
XGBoost / evo2	410	361	0.439	0.859
XGBoost / allele + family + class	410	361	0.910	1.000

blaCTX-M-27

Learner / features	Held out	Above panel	EA	>2 mg/L share
Ridge / allele	170	134	0.047	0.022
Ridge / allele + Evo 2	170	134	0.135	0.575
Ridge / evo2	170	134	0.647	0.948
Ridge / allele + family + class	170	134	0.776	1.000
XGBoost / allele	170	134	0.041	0.022
XGBoost / allele + Evo 2	170	134	0.071	0.791
XGBoost / evo2	170	134	0.576	0.910
XGBoost / allele + family + class	170	134	0.918	1.000

blaCTX-M-55

Learner / features	Held out	Above panel	EA	>2 mg/L share
Ridge / allele	39	36	0.051	0.111
Ridge / allele + Evo 2	39	36	0.103	0.194
Ridge / evo2	39	36	0.872	1.000
Ridge / allele + family + class	39	36	0.744	1.000
XGBoost / allele	39	36	0.000	0.056
XGBoost / allele + Evo 2	39	36	0.231	0.861
XGBoost / evo2	39	36	0.744	1.000
XGBoost / allele + family + class	39	36	0.949	1.000

blaOXA-1

Learner / features	Held out	Above panel	EA	>2 mg/L share
Ridge / allele	218	174	0.862	0.994
Ridge / allele + Evo 2	218	174	0.858	0.994
Ridge / evo2	218	174	0.844	0.994
Ridge / allele + family + class	218	174	0.858	0.994
XGBoost / allele	218	174	0.899	0.994
XGBoost / allele + Evo 2	218	174	0.881	0.994
XGBoost / evo2	218	174	0.748	0.994
XGBoost / allele + family + class	218	174	0.881	0.994

S4. Numerical convergence and computational provenance

The bounded ridge development rerun recorded the following nonconverged fits, counting inner selection and outer final fits together. These counts do not imply that every selected final fit failed to converge.

Population	Features	Nonconverged / total fits
human clinical	Allele	0 / 435
human clinical	Evo 2	0 / 1275
human clinical	Allele + Evo 2	20 / 1275
all sources	Allele	0 / 435
all sources	Evo 2	15 / 1275
all sources	Allele + Evo 2	57 / 1275

For final external prediction (including development-fold selection), the corresponding counts were 7/318 for human-clinical allele + Evo 2, 16/318 for all-source allele + Evo 2, and 3/318 for all-source Evo 2. All other external ridge configurations and both family ridge runs recorded zero nonconverged fits. Full per-fold settings and messages are in `selected_settings.csv`.

Production embedding extraction used four completed H100 tasks, with recorded wall times totaling approximately 3.6 GPU-hours. Software versions and the 64 embedding-shard checksums accompany the release. This count excludes training the foundation model, CPU modelling, genome processing, engineering attempts, and transfers.

Machine-readable supplements

The release includes `full_external.csv`, `final_development.csv`, `all_evaluations.csv` (including superseded development runs, identified by run path), `reference_and_lineage_strata.csv`, `paired_comparisons.csv`, `source_sensitivity.csv`, `variant_results.csv`, and `selected_settings.csv`. Source exclusions re-evaluate existing out-of-fold predictions without retraining. No clinical S/I/R, clinical very-major-error, independent replication, or laboratory-validation result is claimed.